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1 Writing Conventions

In general I use British English.

1.1 Logical rules

Each item in a list will end with a point and start capitalized. If I list something where the order matters like:

1. Step one.
2. Step two.
3. Step three.

and for equivalent properties I use letters: For example, a Morse-Smale function f on a Riemannian manifold (M, g) satisfies:

(a) f is smooth.

(b) The critical points of f are non degenerate.

(c) The (un-) stable manifolds intersect transversal.

1.2 Structure

In this thesis i have the following text-types

(a) Free text that is not in any container.

(b) The theorem container.

(c) The lemma container.

(d) The corollary container.

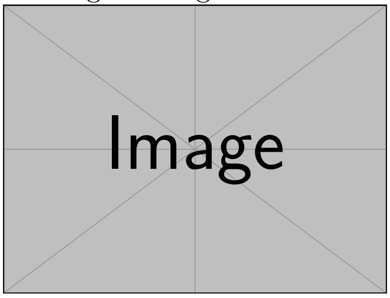
(e) The proposition container.

(f) The prove container.

Free Text is the main narrative of the thesis. It includes explanations, motivations, intuitions, examples, informal remarks, and transitions. Free text connects the formal content and helps guide the reader through the logical structure of the work.

Theorem 1.2.1 (An Example)**.** *Used for major, central mathematical statements that are either original contributions or critical results from the literature. Theorems typically capture the most important claims and are often proved in detail.*

Proof Strategy:
Here I give the geometric idea of the proof and explain the needed steps.



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Lemma 1.2.2 (An Example)**.** *Used for technical results that support the proof of theorems or other key statements. Lemmas are usually not of independent interest but are essential building blocks within a logical argument.*

Corollary 1.2.3 (An Example)**.** Used for statements that follow immediately and straightforwardly from a previously proven theorem or lemma. Corollaries are usually included to highlight direct consequences of major results.

Proposition 1.2.4 (An Example)**.** *Used for results that are mathematically relevant but not as central as theorems. Propositions often encapsulate useful facts, structural properties, or classifications that are too substantial for a lemma but not significant enough for a theorem.*

Proof. This container is pretty obvious □

1.3 Mathemtical Conventions

Elements in a vector bundle will have greek letters. For example $\xi \in TM$. For restrictions I will use the notation $f|_A$ that is written as :

`\restr{f}{A}`.

In the realm of differential geometrie i will **always** use lower indices to denote components. I will **not** use the einstein summation, since it realy appears and I never reallly deal with cumbersome computations (in this context).

1.4 Notation

Symbol	Description	Context
A^t	The transpose of the Matrix A	Linear algebra
M	Smooth manifold	Differential geometry
$\dim M$	Dimension of manifold M	Differential geometry
TM	Tangent bundle of M	Differential geometry
T_pM	Tangent space at point p	Differential geometry
g	Riemannian metric	Riemannian geometry
$\mathrm{Hess}_p(f)$	Hessian of f in p	Morse theory
$M_p(f)$	The matrix correspoding to the Hessian. The used chart is suppressed	Morse theory
$\mathrm{Crit}_k(f)$	Set of critical points of index k	Morse theory
λ_p or λ_p^f	Index of the critical point p	Morse theory

2 Differentiable structures on topological manifolds

This section aims to set the playing field for the thesis. Due to the theory being of intrest for mathematicians and physisists alike there are many (equivalent) definitions used to introduce the objects. The main goal of this section is to avoid confusion, by giving the definitions instead relying on the reader to guess what a certain symbol stands for.

Definition 2.0.1 (Topological Manifold)**.** A second countable Hausdorffspace M is called **topological manifold** of dimension $m \in \mathbb{N}$, if it is locally homeomorphic to $\mathbb{R}^m$. To be precise, if for all $p \in M$ there exists an open neighborhood $U \subseteq M$ of p, an open set $V \subseteq \mathbb{R}^m$ and a map $\varphi : W \to V$ that is a homeomorphism. We call the map $\varphi : U \to V$ a **chart around** p **on** M and φ^{-1} a **local coordinate system around** p **on** M.

Definition 2.0.2 (Differentiable Manifold)**.** Let M be a topological manifold of dimension M.

1. A **differentiable atlas of class** $r \in \mathbb{N} \cup \{\infty\}$ is a family of charts $\mathfrak{A} = (\varphi_i : U_i \to V_i)_{i \in I}$ such that
   a) $\bigcup_{i \in I} U_i = M$, meaning that (U_i) is an open covering of M.
   b) For every pair $(i, j) \in I^2$ the **transition function**:
   $$\begin{aligned} \varphi_{ij} : \varphi_j(U_i \cap U_j) &\to \varphi_i(U_i \cap U_j) \\ x &\mapsto \left(\varphi_i \circ \varphi_j^{-1}\right)(x) \end{aligned}$$
   is differentiable of class r.

   We call such an atlas a C^r-atlas.

2. Two C^r-atlases $\mathfrak{A}$ and $\mathfrak{B}$ are called **equivalent** if the family $\mathfrak{A} + \mathfrak{B} = (\varphi_i, \varphi_j)_{ij}$ is a C^r-atlas.

A **differentiable structure of class** r on M is an equivalence class c of C^r-atlases. For $r = \infty$ we call the pair (M, c) a smooth manifold.

Corollary 2.0.3. Every transition functions φ_{ij} $i, j \in I^2$ of a differentiable atlas $\mathfrak{A} = (\varphi_i)_{i \in I}$ is not just a homeomorphism but also a diffeomorphism due to $\varphi_{ji} = \varphi_{ij}^{-1}$

Definition 2.0.4. Let (M, c) be a differentiable manifold of class r and $U \subseteq M$ open. We call a continuos function

$$f : U \to \mathbb{R}$$

differentiable of class r, if for any one (and hence for all) $(\varphi_i)_{i \in I} = \mathfrak{A} \in c$ the compositions $f \circ \varphi_i^{-1}$ are differentiable of class r. For $r = \infty$ we define:

$$\mathcal{E}(U) = \{f \in U \to \mathbb{R} \text{ continouos} \mid f \text{ is differentiable of class } \infty\}.$$

Corollary 2.0.5. Let (M,c) be a smooth manifold of dimension m and $U \subseteq M$ be a open subset. With pointwise defined operations, the set $(\mathcal{E}(U),+,\cdot,\circ)$ becomes an $\mathbb{R}$–algebra. Furthermore, $\mathcal{E}$ becomes a sheaf of $\mathbb{R}$-algebras.

Proof. There is not really a need for a proof. However, it might help to work through the definition of a sheaf as a reminder. First, $\mathcal{E}$ is a presheaf, where the restriction in the domain of a function gives the needed restriction homomorphism:

$$\begin{aligned}\mathrm{res}_V^U : \mathcal{E}(U) &\to \mathcal{E}(V)\\ f &\mapsto f|_V\,.\end{aligned}$$

The required properties of a presheaf are trivial. Furthermore, this gives a sheaf as the requirement of locality is trivial for functions and the property of gluing is also trivial for functions, since differentiability is a local property. □

Definition 2.0.6. If $p \in M$ is fix, $f \in \mathcal{E}(U)$ and $g \in \mathcal{E}(U')$ such that $p \in U \cap U'$ we say that f and g have the same **germ in** p, if there is another open neighborhood $W \subseteq U \cap U'$ of p such that $f|_w = g|_W$. This defines an equivalence relation $\sim_p$. An equivalence class s of local functions around p is called a **germ in** p. We write $s = f_p$, if $s = [f]$ with $f \in \mathcal{E}(U)$. We write

$$\mathcal{E}_p(M) = \left(\sum_{U \text{open}, p \in U} \mathcal{E}(U)\right) \Big/ \sim_p\,.$$

For the set of germs and call it the **stalk in** p. Here $\sum$ denotes the co-product (also called sum) in **Top** and hence the disjoint union.

Corollary 2.0.7. For a smooth manifold (M,c) the set $\mathcal{E}_p(M)$ inherits an $\mathbb{R}$–algebra structure from the $\mathcal{E}(U)$. Furthermore, it carries a natural (evaluation-)homomorphism:

$$\begin{aligned}\mathrm{eval}_p : \mathcal{E}_p(M) &\to \mathbb{R}\\ f_p &\mapsto f(p) =: f_p(p)\end{aligned}$$

The stalks are also local rings with maximal ideal $\mathfrak{m}_p = \ker(\mathrm{eval}_p)$. Hence, the pair $(M,\mathcal{E})$ gives us a locally ringed space.

Proof. Here, we only need to prove the statement about the locality of the stalks. This follows from $f_p \in \mathcal{E}_p(M)$ being invertible if and only if $f(p) \neq 0$ which is the same as $f_p \notin \ker(\mathrm{eval}_p)$. □

Definition 2.0.8. Let (M,c) be a smooth manifold of dimension m and $p \in M$. We call an $\mathbb{R}$-linear map $\delta : \mathcal{E}_p(M) \to \mathbb{R}$ a **derivation**, if it satisfies the Leibnitz-rule:

$$\delta(f_p \cdot g_p) = \delta(f_p) g_p(p) + f_p(p)\delta(g_p) \quad \text{for all } f,g \in \mathcal{E}_p(M)\,.$$

We call $\mathrm{Der}_{\mathbb{R}}(\mathcal{E}_p(M),\mathbb{R})$ the set of derivations and give it a $\mathbb{R}$ vector space structure by pointwise operations. We define the **tangent space of** M **at** p to be the vector space

$$TM_p := \mathrm{Der}_{\mathbb{R}}(\mathcal{E}_p(M),\mathbb{R})\,.$$

Corollary 2.0.9. Let (M,c) be a smooth manifold and $\varphi : U \to V$ be a chart around p with $x_0 = \varphi(p)$ $(\varphi \in \mathfrak{A} \in c)$. Then

$$\xi = \frac{\partial}{\partial x_j}\Big|_p : \mathcal{E}_p(M) \to \mathbb{R}$$
$$f_p \mapsto \xi(f_p) = \frac{\partial}{\partial x_j}\Big|_{x_0} (f \circ \varphi^{-1})$$

is well-defined and a tangent vector. In fact, the family

$$\left(\frac{\partial}{\partial x_1}\Big|_p, \ldots, \frac{\partial}{\partial x_m}\Big|_p\right)$$

defines a basis of TM_p. Hence, the dimension of TM_p is m.

Definition 2.0.10. For a smooth manifold (M,c) the sum $\sum_p TM_p$ comes with a natural projection

$$\pi : TM \to M$$
$$\xi \mapsto p \text{ where } \xi \in TM_p$$

Furthermore, the **local vector fields** with respect to a chart $\varphi : U \to V$

$$\frac{\partial}{\partial x_j} : U \to \pi^{-1}(U)$$
$$p \mapsto \frac{\partial}{\partial x_j}\Big|_p$$

induce a local trivialization:

$$\pi^{-1}(U) \cong U \times \mathbb{R}^m \,.$$

We can induce a topology on TM such that all those trivializations are continuos. This then gives an atlas for TM such that we have a $2m$-dimensional manifold. To be precise, the atlas is given by the maps $\pi^{-1}(U_i) \to R^m \times R^m; x \mapsto (\pi(x), q_{\varphi\circ\pi(x)}(x))$ where q_p denotes the coordinate map corresponding to the basis $(\frac{\partial}{\partial x_1}\big|_p, \ldots, \frac{\partial}{\partial x_m}\big|_p)$ that depends on the chart φ_i. In fact, this yields a smooth manifold and a (smooth) vector bundle of dimension m. We call TM the **tangent bundle**.

Definition 2.0.11 (The Derivative)**.** Let (M,c) and (M',c') be smooth manifolds (from now on we suppress the differentiable structure in our notation). We call a continuous function $f : M \to M'$ **smooth**, if for all $\varphi \in \mathfrak{A} \in c$ and $\varphi' \in \mathfrak{A}' \in c'$ the maps

$$\varphi' \circ f \circ \varphi^{-1} : V \to V'$$

are smooth. A given smooth function induces a smooth function between the Tangent bundles as follows:

$$Df : TM \to TM' \;, Df_p(\xi)(g_p) = \xi((g \circ f)_p)$$

Here, $\xi \in T_pM$, $g_p \in E_p(M')$.

Corollary 2.0.12. Let $f : M \to \mathbb{R}$ be smooth and $\varphi : U \to V$ be a chart. Then we can interprete $\mathrm{d}f$ as a one-form. To be prezice asume q to be the coordinate funktion $T\mathbb{R} \to \mathbb{R}$ from the basis induced by the identity as a chart. $v \in \Gamma TM$ we have

$$q \circ \mathrm{d}f(v) = v \cdot f$$

Here on the righd side, f denotes a map $p \mapsto f_p$ such that $v(f)(p) := v_p(f_p)$ is well defined. We keep this notations so vector fields can take funktions as an input.

Proof. We proof this by showing it for the section $\frac{\partial}{\partial x_i}$ and thereby for any, since those sections form a basis of the space of sections as a $C^\infty(M,\mathbb{R})$ vectorspace. Now let $g_p \in \mathcal{E}_p(\mathbb{R})$ and $\varphi(p) = x_0$. Then

$$\mathrm{d}f(\frac{\partial}{\partial x_i}\Big|_p)(g_p) = \frac{\partial}{\partial x_i}\Big|_p(g \circ f)_p = \frac{\partial}{\partial x_i}\Big|_{x_0}(g \circ f \circ \varphi^{-1}) = \frac{\partial}{\partial x}\Big|_p(g_p) \cdot \frac{\partial}{\partial x_i}\Big|_p(f_p)$$

Hence, $\mathrm{d}f(v) = v(f)\frac{\partial}{\partial x}$ letting us conclude the statement. □

Definition 2.0.13 (A Metric)**.** Let M be a smooth manifold. A section $g \in \Gamma(TM^* \otimes TM^*)$ into the tensor product of the dual Tangend bundle with itself is a **riemannian metric** if the following are satisfied for all $p \in M$:

1. g is **non degenerate**, meaning for a $v_p \in T_pM$ $g_p(v_p, w_p) = 0$ $\forall w_p \in T_pM$ then $v_p = 0$.
2. g is **symmetric**, meaning $g_p(v_p, w_p) = g_p(w_p, v_p)$ $\forall v_p, w_p \in T_pM$.
3. g is **positiv definite**, meaning that $g_p(v_p, v_p) \geq 0 \forall v_p$ and vanishes only for $v_p = 0$.

We cal a touple (M, g) a **riemannian manifold**.

Definition 2.0.14. Let M, N, L be smooth manifolds and $f : M \times N \to M' \times N'$. Let $X \in \gamma(TM)$ and $Y \in \Gamma(TN)$. We define $\tilde{X} \in \Gamma(T(M \times N))$ as follows. Let $f_p \in \mathcal{E}_p(M, N)$

$$\tilde{X}(f_p) = X((f \circ i_M)_{p_M}) \quad \text{where } p_M = \pi_1(p)\,.$$

Definition 2.0.15 (The Gradient)**.** Assume that (M, g) is a smooth manifold together with a riemannian metric. Let $f : M \to \mathbb{R}$ be a smooth function. We define its **gradient** to be a vector field $\mathrm{grad}(f) \in \Gamma(TM)$ such that for any vecor field V:

$$q \circ \mathrm{d}f(V) = g(\mathrm{grad}(f), V)\,.$$

Here q denotes the canonical coordinate funtion $T\mathbb{R} \to \mathbb{R}$.

Definition 2.0.16. Let (M, g) be a riemannian manifold and $\varphi : U \to V$ be a chart. We define the smooth functions $g_{ij} : U \to \mathbb{R}$ to be :

$$g_{ij} = g(\frac{\partial}{\partial x_i}, \frac{\partial}{\partial x_j})\,.$$

Then if two vectorfields v, w over U are given with $v = \sum v^i \frac{\partial}{\partial x_i}$ and $w = \sum_i w^i \frac{\partial}{\partial x_i}$ we can calculate the metric locally:

$$g(v,w) = \sum_{ij} g_{ij} v^i w^j \ : U \to \mathbb{R}$$

Furthermore, we can invert the matricies $(g_{ij}(p))_{ij}$ for all p (which is a smooth procedure which can be seen by the construction via cramers rule) to get smooth functinos

$$\begin{aligned} g^{ij} : U &\to \mathbb{R} \\ p &\mapsto g^{ij}(p) = \left((g_{kl}(p))_{kl}^{-1}\right)_{ij} \end{aligned}$$

Lemma 2.0.17. *Let $\varphi : U \to V$ be a chart. Then the gradient has the local form:*

$$\mathrm{grad}(f) = \sum_{i,j} g^{ij} \left(\frac{\partial}{\partial x_i}(f)\right) \frac{\partial}{\partial x_j} =: \sum_{i,j} g^{ij} \frac{\partial f}{\partial x_i} \frac{\partial}{\partial x_j}\,.$$

Proof. Assume that $v, w \in \Gamma(TM)$ such that $v = \sum_i v^i \frac{\partial}{\partial x_i}$ and $w = \sum_i w^i \frac{\partial}{\partial x_i}$. Then $g(v,w) = \sum_{i,j} g_{ij} v^i w^i$. Suppose that $\mathrm{grad}(f) = \sum_j G^j \frac{\partial}{\partial x_j}$ and q is the coordinate map $T\mathbb{R} \to \mathbb{R}$ in each tangend space. Then by definition of the gradient we have:

$$\frac{\partial}{\partial x_j}(f) = q \circ \mathrm{d}f(\frac{\partial}{\partial x_j}) = q \circ \frac{\partial f}{\partial x_j} = g(\mathrm{grad}(f), \frac{\partial}{\partial x_j}) = \sum_{ij} g_{ij} G^i$$

Hence,

$$\begin{aligned} (G^1, \cdots G^m)(g_{ij}) &= \left(\frac{\partial}{\partial x_1}(f), \cdots, \frac{\partial}{\partial x_m}(f)\right) \\ \Leftrightarrow (G^1, \cdots G^m) &= \left(\frac{\partial}{\partial x_1}(f), \cdots, \frac{\partial}{\partial x_m}(f)\right)(g^{ij})\,. \end{aligned}$$

But then we can conclude that $G^j = \sum_i g^{ij} \frac{\partial}{\partial x_i}(f)$.

□

Definition 2.0.18 (A Dynamical System)**.** Let M be a smooth manifold. A **dynamical system on** M is given by a smooth map $\varphi : \Omega \to M$ such that:

1. $\{0\} \times M \subseteq \Omega \subseteq \mathbb{R} \times M$ open, and for any $p \in M$ the set $I(p) := \pi_1(\Omega \cup \mathbb{R} \times \{p\})$ is connected. (π_1 is the projection onto the first factor)
2. For all $p \in M$ and $t \in I(p)$ we have that $s \in I(\varphi(t,p))$ if and only if $s + t \in I(p)$. Then we have:

$$\varphi(s, (\varphi(t,p))) = \varphi(s+t, p)$$

3. For all p we require $\varphi(0, p) = p$.

We will write $\varphi : t(p) := \varphi(t, p)$ to keep the notation bearable. If for all p $I(p) = \mathbb{R}$ we call the system **global**.

Corollary 2.0.19. For a global dynamical system φ on a smooth manifold M the map

$$\rho : \mathbb{R} \to \mathrm{Diff},\ t \mapsto \varphi_t$$

is a homomorphism. To see the well definition notice how for all $t \in \mathbb{R}$ φ_{-t} is an inverse of φ_t.

Definition 2.0.20 (Vector Field of a Dynamical System)**.** Let φ be a dynamical system on a smooth manifold M. Then we call the mapping

$$p \mapsto X(p) := \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} \varphi_t(p) \in T_pM$$

the **associated vecor field**. This is in fact a smooth vector field.

Whilst differentiating leads to a smooth vector field, we can also go the other direktion by locally integrating a vector field to get a flow. To be precise: Those concepts are inverse to one another or in other words

hier genauer werden

Theorem 2.0.21. *Let M be a smooth manifold. Then there is a one to one correspondence between global maximal flows and vector fields.*

3 Morse Theory

For everything concerning Morse theory our main reference will be the wonderful book [**banyaga2004lectures**]. In Morse Theory, we will always deal with asmooth funkition $f : M \to \mathbb{R}$ from a smooth Riemannian manifold (M, g) that is also compact. We will denote its dimension by $m = \dim(M)$.

Definition 3.0.1 (Kritical Points)**.** We call a point $p \in M$ **critical**, if $\mathrm{d}f|_p$ vanishes meaning that

$$\mathrm{d}f|_p : T_pM \to T_{f(p)}\mathbb{R}$$

is trivial. By $\mathrm{Crit}(f)$ we denote the **set of critical points**.

Definition 3.0.2 (The Hessian)**.** Let $f : M \to \mathbb{R}$ be a smooth funktion. The **Hessian** of f at $p \in \mathrm{Crit}(f)$ is a symmetric bilinear funktion

$$\mathrm{Hess}_p(f) : T_pM \times T_pM \to \mathbb{R}$$

that is defined as follows. Let $\xi_p, \zeta_p \in T_pM$ be two tangend vectors. Now choos two vectorfields Ξ and Z in a neighbourhoof of p such that $\Xi(p) = \xi_p$ and $Z(p) = \zeta_p$. With this define the Hessian to be:

$$\mathrm{Hess}_p(f)(\xi_p, \zeta_p) = (\Xi \cdot (Z \cdot f))\,(p)$$

Compare 2.0.12 for the notation.

Definition 3.0.3. A smooth function $f : M \to \mathbb{R}$ is called **Morse**, if for any critical point.

The Hessian is a Symmetric bilinear and non-degnereate Map. For such a map there exists the following theorem:

Theorem 3.0.4 (Sylvesters Law of Inertia)**.** *Let $A \in \mathrm{Mat}(n, \mathbb{R})$ be symmetric and let $T, T' \in \mathrm{GL}(n, \mathbb{R})$ and $k, k', l, l' \in \mathbb{N}$ such that*

$$T^t \circ A \circ T = \begin{pmatrix} \mathbb{1}_k & 0 & 0 \\ 0 & -\mathbb{1}_l & 0 \\ 0 & 0 & 0_{n-k-l} \end{pmatrix} \text{ and } T'^t \circ A \circ T' = \begin{pmatrix} \mathbb{1}_{k'} & 0 & 0 \\ 0 & -\mathbb{1}_{l'} & 0 \\ 0 & 0 & 0_{n-k'-l'} \end{pmatrix}$$

then $k = k'$, $l = l'$ and $\mathrm{rank}(A) = k + l$.

Proof. Since T, T' are invertible we have that

$$k + l = \mathrm{rank}(T^t \circ A \circ T) = \mathrm{rank}(A) = \mathrm{rank}(T'^t \circ A \circ T') = k' + l' \,. \tag{1}$$

Hence it suffices to show that $k = k'$. Which we will do by proofing the claim:

$$k = \max\left\{\dim(U) | U \subseteq \mathbb{R}^n \text{subspace such that } x^t A x > 0 \ \forall x \in U \setminus \{0\}\right\}$$

So we start by showing „≤“: Denote the first k colomus of T with $x_1, ..., x_k$. They form a basis of $\mathbb{R}^n$ and with $0 \neq x = \sum_{i=1}^k \lambda i x_i$ we have that by bilinearity

$$x^t A x = \sum_{i=1}^k \lambda_i x_i^t A x = \sum_{i,j=1}^k \lambda_i \lambda_j x_i^t A x_j = \sum_{i=1}^k (\lambda_i)^2 \geq 0\,. \tag{2}$$

This conculdes the first inequality.

Now let U be any k−dimensional subspace such that for all non zero $x \in U$ $x^t A x > 0$. By a calculation analog to the one above we have that for any $x \in W := \operatorname{span}(x_{k+1}, \ldots x_n)$ the number $x^t A x$ is less or equal to zero. Hence, $W \cap U = \{0\}$ and with this we conclude:

$$\dim(U) = \dim(U + W) - \dim(W) + \dim(U \cap W) \leq (k + (n - k)) - (n - k) + 0 = k\,. \tag{3}$$

This is the last inequality proving the statement. □

Corollary 3.0.5. Let $\varphi : U \to V$ be a chart around a critical point $p \in M$. Then in said chart the matrix corrsponding to the Hessian has the form:

$$M_p(f) = \left(\frac{\partial^2 (f \circ \varphi^{-1})}{\partial x_i \partial x_j}\right)_{i,j}$$

Proof. This is just an easy calculation, since $\operatorname{Hess}_p(f)\left(\frac{\partial}{\partial x_i}\Big|_p, \frac{\partial}{\partial x_j}\Big|_p\right)$ is given by

$$\left(\frac{\partial}{\partial x_i}\Big|_p \left(\frac{\partial}{\partial x_j}(f)\right)_p\right) = \left(\frac{\partial^2 (f \circ \varphi^{-1})}{\partial x_i \partial x_j}\right)$$

□

The Hessian matrix $M_p(f)^i$ in two different charts $\varphi_i : U_i \to V_i$ for $i = 1, 2$ transforms via a congruence transformation by the Jacobian of the transition function If we have two different charts. Then the Matrix $M_p(f)$ of the Hessian transformes from one base to the other via conjugation of the differential of the transition function $\varphi_{12} := \varphi_1 \circ \varphi_2^{-1}$:

$$M_p^1(f) = \mathrm{d}\varphi_{12}^t \cdot M_p^2(f) \cdot \mathrm{d}\varphi_{12}$$

Hence we can define the following:

Definition 3.0.6. By Sylvesters Law of interta, the number l that denotes the dimension of the maximal subspace on which a matrix is negative definite, is a invariant under conjugation. We call that number the index of said matrix. Now in the setting of Morse theory we assign to each kritical point p the index of $M_p(f)$ and call that the **(Morse-)index** of p, denoted with λ_p. If we cannot suppres the function without loss of clearity we will write λ_p^f. We denote the set of all critical points of index k by $\operatorname{Crit}_k(f)$.

Theorem 3.0.7 (Morse Lemma)**.** *Let p be a critical point of index k of the Morse funktion $f : M \to \mathbb{R}$. There exists a centered chart $\varphi : U \to \mathbb{R}^m$ arround p such that*

$$(f \circ \varphi^{-1})(x_1, \dots, x_m) = f(p) + \sum_{i=1}^{k} -x_i^2 + \sum_{i=k+1}^{m} x_i^2$$

Proof. Compar for example Lemma3.11 in [**banyaga2004lectures**]. ☐ hier noch mehr Quellen!

Definition 3.0.8. Let $\varphi_t : M \to M$ be the local one-parameter group of diffeomorphisms generated by $-\mathrm{grad}(f)$, i.e. if $e : \mathbb{R} \to T\mathbb{R}$ denotes the canonical unit vector field that comes from the identity as the global chart on $\mathbb{R}$, we have that

$$\frac{\mathrm{d}}{\mathrm{d}t}\varphi_t(x) \circ e = -\mathrm{grad}(f)(\varphi_t(x))\,, \quad \text{and}$$
$$\varphi_0(x) = x\,.$$

We call that group the **gradient vevtor field** of f with respect to the gradient g. The integral curve $\gamma_x : (a,b) \to M$ given by

$$\gamma_x(t) = \varphi_t(x)$$

is called a gradient **flow line**.

Next we will deepen our **local** understanding of said vector fiel. Notice how the Morse Lemma is a statement about smooth manifolds-no metric needed nor mentioned. In fact there are many Morse charts and we can finde one, where the riemannian metric in p is diagonal with non-zero. To proof such a statement we first need a somewhat obvious lemma:

Lemma 3.0.9. *Assume that $\psi : U \to V$ is a chart and $L : R^m \to R^m$ a linear function. Denote the local coordinate maps of the Tangend bundle in a point p that comes from a basis in said point induced from a chart χ by q_χ. Then the diagramm*

$$\begin{tikzcd}
T_pM \arrow[r, "q_\psi|_p"] \arrow[dr, "q_{L\circ\psi}|_p"'] & \mathbb{R}^m \arrow[d, "L"] \\
& \mathbb{R}^m
\end{tikzcd}$$

commutes for all $p \in U$.

Proof. We show this by showing the inverse: For any $v \in \mathbb{R}^m$ we have the two $(q_\psi\big|_p)^{-1} \circ L^{-1}(v)$ and $(q_{\psi\circ L}\big|p)^{-1}$ and claim that they are the same. Assume that $L^{-1} = (\lambda_{ij})_{ij}$, $\psi(p) = x_0$, $L(x_0) = \tilde{x_0}$, $(v_1, \dots v_m) = v \in \mathbb{R}^m$ and denote the basis of T_pM coming from the chart ψ by $\left(\frac{\partial}{\partial x_i}\big|_p\right)_i$ and the one coming from $L^{-1} \circ \psi$ by $\left(\frac{\partial}{\partial \tilde{x}_i}\big|_p\right)_i$. We now calculate for $f_p \in \mathcal{E}_p M$:

$$\begin{aligned}
\left[(q_{\psi\circ L}\big|_p)^{-1}(v)\right](f_p) &= \sum_i v_i \frac{\partial}{\partial \tilde{x}_i}\Big|_p (f_p) \\
&= \sum_i v_i \frac{\partial}{\partial x_i}\Big|_{\tilde{x}_0} (f \circ \psi^{-1} \circ L^{-1}) \\
&= \sum_i v_i \sum_j \frac{\partial f \circ \psi^{-1}}{\partial x_j}\Big|_{x_0} \cdot \frac{\partial L_j^{-1}}{\partial x_i}\Big|_{\tilde{x_0}} \\
&= \sum_{i,j} v_i \frac{\partial}{\partial x_j}\Big|_p (f_p) \cdot \lambda_{ji} \\
&= \sum_{i,j} v_i q_\psi\big|_p^{-1}(e_j)(f_p) \cdot \lambda_{ji} \\
&= \left[q_\psi\big|_p^{-1}\Big(\sum_{i,j} v_i \lambda_{ji} e^j\Big)\right](f_p) \\
&= \left[q_\psi\big|_p^{-1}(L^{-1}(v))\right](f_p)
\end{aligned}$$

This concludes the proof. □

Now we can proof that a Morse chart exists where the gradient is of particular easy form:

Theorem 3.0.10 (Simultaneous Diagonalisation of Quadratic Forms)**.** *Let p be a critical point of a Morse function f on a manifold M and let $g(p)$ be the Riemannian metric on T_pM. Then there exists a Morse chart ψ arround p such that the representation of $g(p)$ in the basis induced by the Morse chart is given by a a diagonal matrix $diag(\mu_1, \ldots \mu_m)$ with $\mu_i > 0$ for all i .*

Proof. The quadratic form of $f \circ \psi^{-1} - f(p)$ in the coordinates of any Morse chart ψ is $q_H(v) = v^T H v$. The quadratic form induced by the Riemannian metric $g(p)$ is $q_G(v) = v^T G v$, where $v \in \mathbb{R}^m$ are the coordinate vectors with respect to the Morse chart. To be precice this means for $v, w \in \mathbb{R}^m$:

$$g \circ (q_\psi\big|_p \oplus q_\psi\big|_p)(v, w) = v^T G w \text{ and } f \circ \psi^{-1}(v) = f(p) + v^T H v\,.$$

We now aim to manipulate ψ such that G becomes diagonal: Since $g(p)$ is positive definite, G is also positive definite, and H is symmetric.

Consider the generalized eigenvalue problem $Hv = \lambda G v$. Since H and G are real symmetric matrices and G is positive definite, this problem has m real eigenvalues $\lambda_1, \ldots, \lambda_m$ and corresponding eigenvectors $v_1, \ldots, v_m$, which can be chosen to be orthogonal with respect to the bilinear form defined by G, such that $v_i^T G v_j = \delta_{ij}$, since if v_i and v_j are different eigenvectors with respect to different eigenvalues, we can calculate:

$$v_j^T H v_i = (Hv_j)^T v_i = \lambda_j (Gv_j)^T v_i = \lambda_j v_j^T G(v_i) \quad \text{and} \quad v_j^T H v_i = v_j^T \lambda_i G(v_i) = \lambda_i v_j^T G(v_i)\,.$$

Hence we have the equality

$$\lambda_j (Gv_j)^T v_i = \lambda_i v_j^T G(v_i) \Leftrightarrow \underbrace{(\lambda_j - \lambda_i)}_{\neq 0} v_j^T G(v_i) = 0$$

Hence all eigenspaces are orthogonal and we can use Gram Schmitt to make the bases of the eigenspaces orthogonal and by rescaling orthonormal.

Let $L = [v_1|\dots|v_m]$ be the matrix whose columns are these G-orthonormal eigenvectors. The linear change of coordinates $y = Lz$ leads to new coordinates z. In these new coordinates, the quadratic forms transform as follows:

$$\begin{aligned} q_G(y) &= y^T G y = (Lz)^T G(Lz) = z^T L^T G L z = z^T I z = \sum_{i=1}^m z_i^2 \\ q_H(y) &= y^T H y = (Lz)^T H(Lz) = z^T L^T H L z \end{aligned}$$

Since $Hv_i = \lambda_i G v_i$, we have $L^T H L = \operatorname{diag}(\lambda_1, \dots, \lambda_m)$. The eigenvalues λ_i are real and have the same signature as the eigenvalues of H (l negative, $m - l$ positive).This is true to sysvesters law of inertia. By a further scaling of the v_i and a reordering the matrix $L^T H L$ can be brought into the form $\operatorname{diag}(-1, \dots, -1, 1, \dots, 1)$. (by doing this however, the matrix $L^T G L$ becomes $L^T G L = \operatorname{diag}(\mu_1, \dots \mu_m)$ with $\mu_j > 0\ \forall j$). If ψ is the Morse chart we started with, then $L^{-1} \circ \psi$ is the new Morse chart:

$$f \circ (\psi^{-1} \circ L)(y) = f \circ \psi^{-1}(L(y)) = q_H(L(y) = \sum_{i=1}^{l} -y_i^2 + \sum_{i=l+1}^{m} y_i^1$$

Furthermore, we want to inspect the metric $g\big|_p : T_p M^2 \to \mathbb{R}$ with respect to the chart $L^{-1} \circ \psi$-induced basis $\left(\frac{\partial}{\partial x_1}\big|_p, \dots, \frac{\partial}{\partial x_m}\big|_p\right)$. By the lemma 3.0.9

$$g(q_{L^{-1}\circ\psi}\big|_p^{-1}, q_{L^{-1}\circ\psi}\big|_p^{-1})(v, w) = g(q_\psi\big|_p^{-1}, q_\psi\big|_p^{-1})(Lv, Lw) = (Lv)^T G L W = v^T L^T G L w$$

□

Definition 3.0.11. We call a Morse chart where the Matrix corresponding to the gradient in the critical point is of diagonal form a **refined Morse chart**.

Using such a chart, we can start looking at the flow itself and try to find easy local formulars around critical points. For this we start by looking at the Jacobian of the gradient in a critical point:

Definition 3.0.12 (Jacobian of the Gradient)**.** The gradient is a section into the tangent bundle, $\operatorname{grad}(f) : M \to TM$. Let $\psi : M \supseteq U \to V \subseteq \mathbb{R}^m$ be a chart. The coordinate map q_ψ on TU assigns to a vector $v \in T_pM$ (where $p \in U$ with $\psi(p) = x$) its components with respect to the basis $\left(\frac{\partial}{\partial x_i}\big|_p\right)$, i.e., $q_\psi(v) = (v_1, \ldots, v_m)$ if $v = \sum_{i=1}^m v_i \frac{\partial}{\partial x_i}\big|_p$. We define the Jacobian of the gradient with respect to the chart ψ as the Jacobian matrix of the coordinate representation of the gradient in this chart:

$$J(\operatorname{grad}(f))_\psi(x) := J\left(q_\psi \circ \operatorname{grad}(f) \circ \psi^{-1}\right)(x) = \left(\frac{\partial (q_\psi \circ \operatorname{grad}(f) \circ \psi^{-1})_i}{\partial x_j}(x)\right)_{ij}$$

Lemma 3.0.13. *Let ψ be a coordinate system arround a critical point p. I.e. $\psi : p \in U \to V$ with $\psi(p) = 0$ and let $\left(\frac{\partial}{\partial x_1}\big|_x, \ldots, \frac{\partial}{\partial x_m}\big|_x\right)$ be the induced basis of T_xM Then the Jacobian of the gradient reads*

$$J(\operatorname{grad}(f))_\psi(0) = \left(\sum_k g^{ki}(0) \frac{\partial^2 f \circ \psi^{-1}}{\partial x_j \partial x_k}\right)_{ij} . \tag{4}$$

Proof. Let q_ψ denote the coordinate funktion $TU \to R^m$ induced from the basis $\left(\frac{\partial}{\partial x_1}\big|_x, \ldots, \frac{\partial}{\partial x_m}\big|_x\right)$. then the gradient in ψ reads:

$$q_\psi \circ \operatorname{grad}(f) = \left(\sum_k g^{k1} \frac{\partial f}{\partial x_k}, \ldots, \sum_k g^{km} \frac{\partial f}{\partial x_k}\right)$$

Hence, we can differentiate:

$$\begin{aligned} \frac{\partial}{\partial x} q_\psi \circ \operatorname{grad}(f) \circ \psi^{-1} &= \left(\frac{\partial}{\partial x_j} \sum_k g^{ki} \frac{\partial f \circ \psi^{-1}}{\partial x_k}\right)_{ij} \\ &= \left(\sum_k \left[\frac{\partial g^{ki}}{\partial x_j} \frac{\partial f \circ \psi^{-1}}{\partial x_k} + g^{ki} \frac{\partial^2 f \circ \psi}{\partial x_j\, \partial x_k}\right]\right)_{ij} . \end{aligned}$$

and now in $p = \psi^{-1}(0)$ we have

$$\frac{\partial}{\partial x} q_\psi \circ \operatorname{grad}(f) \circ \psi^{-1}\Big|_0 = \left(\sum_k \left[g^{ki}(0) \frac{\partial^2 f \circ \psi^{-1}}{\partial x_j\, \partial x_k}\right]\right)_{ij} .$$

□

Now that we now how the gradient locally looks like ist reasonable to assume, that the flow locally looks like its exponential:

Lemma 3.0.14. *Let $t \in \mathbb{R}$ be small enough and $\varphi_t : M \to M$ be the flow map corresponding to the negative gradient. Assume that U is the domain of a chart ψ and $p \in U$ such that $\varphi_t(p) \in U$. Then the linear map $\mathrm{d}\varphi_t\big|_p : T_pM \to T_{\varphi_t(p)}M$ has a local representation in the coordinates induced by ψ given by:*

$$\frac{\partial}{\partial x}\psi \circ \varphi_t(\psi^{-1}(x)) = \exp\left(-\left(\sum_k g^{ki}(0)\frac{\partial^2 f \circ \psi^{-1}}{\partial x_j \partial x_k}\right)_{ij} \cdot t\right).$$

Proof. For this we consider the function $(t,x) \mapsto \psi \circ \varphi_t(\psi^{-1}(x)) : \mathbb{R} \times U \to U$ For fixed x we can calculate

$$q_\psi \frac{\mathrm{d}}{\mathrm{d}t}\varphi_t(\psi^{-1}(x)) = -q_\psi \circ \operatorname{grad}(f)(\varphi_t(\psi^{-1}(x)))$$

and by changing the order of integration we have:

$$\begin{aligned}
\frac{\partial}{\partial t}\frac{\partial}{\partial x}\psi \circ \varphi_t(\psi^{-1}(x)) &= \frac{\partial}{\partial x}\frac{\partial}{\partial t}\psi \circ \varphi_t(\psi^{-1}(x)) \\
&= \frac{\partial}{\partial x} q_\psi \frac{\mathrm{d}}{\mathrm{d}t}\varphi_t(\psi^{-1}(x)) \\
&= -\frac{\partial}{\partial x} q_\psi \circ \operatorname{grad}(f)(\varphi_t(\psi^{-1}(x))) \\
&= -\frac{\partial}{\partial x} q_\psi \circ \operatorname{grad}(f)\left(\psi^{-1} \circ \psi \circ \varphi_t(\psi^{-1}(x))\right) \\
&= -\frac{\partial}{\partial x}\left(q_\psi \circ \operatorname{grad}(f) \circ \psi^{-1}\right)\frac{\partial}{\partial x}\left(\psi \circ \varphi_t(\psi^{-1}(x))\right)
\end{aligned}$$

Hence by the theory of linear differential equation with $\frac{\partial}{\partial x}\psi \circ \varphi_0(\psi^{-1}(x)) = \mathbb{1}_m$ we have:

$$\frac{\partial}{\partial x}\psi \circ \varphi_t(\psi^{-1}(x)) = \exp\left(-\frac{\partial}{\partial x}\left(q_\psi \circ \operatorname{grad}(f) \circ \psi^{-1}\right) \cdot t\right).$$

The preceeding lemma now finishes the proof.

□

Hence in a refind Morse chart we have:

Remark 3.0.15. Let ψ be a refined Morse chart, and hence:

(a) $g^{ik}(0) = \delta_{ik}\mu_k$ with $\mu_k > 0$ for any k.

(b) $\frac{\partial^2 f \circ \psi^{-1}}{\partial x_j \partial x_k} = s_k 2\delta_{jk}$ where $s_k = -1$ for $k \leq \lambda_p$ and $s_k = 1$ else.

Then the Jacobian of the flow in the critical point p reads:

$$\begin{aligned}
\frac{\partial}{\partial x}\psi \circ \varphi_t(\psi^{-1}(x)) &= \exp(-J(\mathrm{grad}(f))(p)\cdot t) \\
&= \exp\left(-\sum_k \underbrace{g^{ki}}_{=\delta_{ik}\mu_k}(0)\underbrace{\frac{\partial^2 f \circ \psi^{-1}}{\partial x_j \partial x_k}}_{=s_k 2\delta_{jk}} \cdot t\right) \\
&= \mathrm{diag}\left(e^{2\mu_1 t}, \dots, e^{2\mu_{\lambda_p} t}, -e^{2\mu_{\lambda_p+1} t}, \dots, -e^{2\mu_m t}\right)
\end{aligned}$$

4 Morse-Theoretic Atiyah-Hirzebruch Spectralsequence

Definition der Conley Filtration

The Main statement now is that the connecting homomorphism of K-Theory and of the long exact sequence given by a special Filtration in Morse Theory are comparable:

Theorem 4.0.1. *Given the filtration $N_{-1} \subseteq N_0 \subseteq \cdots \subseteq N_{m-1} \subseteq N_m = M$ from **??**. The following diagram commutes:*

$$\begin{tikzcd}
C^k(M, \mathfrak{A}, \tilde{K}^l(pt)) \arrow[r, "\partial^p"] \arrow[d] & C^{k+1}(M, \mathfrak{A}, \tilde{K}^l(pt)) \arrow[d] \\
\bigoplus_{p \in \mathrm{Crit}_k(f)} \tilde{K}^{k+l}(N_p, L_p) \arrow[r, "\Delta_{k+l}"] \arrow[d] & \bigoplus_{q \in \mathrm{Crit}_{k+1}(f)} \tilde{K}^{k+l+1}(N_q, L_q) \arrow[d] \\
K^{k+l}(N_k, N_{k-1}) \arrow[r, "\delta_{triple}"] & K^{p+l+1}(N_{k+1}, N_k)
\end{tikzcd}$$

Corollary 4.0.2. Lets say that $p \in -\mathbb{N}$ is negative for the moment. We have the three spaces

$$A := N_q \cup N_p \quad , \quad B := L_q \cup N_p \quad , \quad C := L_p \cup (L_q \setminus \mathring{N}_p)\,. \tag{5}$$

By definition we get the triple connecting morphism as the composition of the inclusion with the connecting homomorphism of the pair::

$$\begin{tikzcd}
 & K^p[N_q \cup N_p, L_q \cup N_p] \arrow[r, Rightarrow] & K^p[N_q \cup N_p \cup /(L_q \cup N_p)] \\
 & & K^p[N_q \cup N_p \cup C_1(L_q \cup N_p)] \arrow[u, "(m^*)^{-1}"] \\
K^{p-1}[L_q \cup N_p] \arrow[uur, "\delta_{pair}"] \arrow[r, Rightarrow] & K^p[S_1 \wedge (L_q \cup N_p)] \arrow[r, "\alpha^*"] & K^p[N_q \cup N_p \cup C_1(L_q \cup N_p) \cup C_2(N_q \cup N_p)] \arrow[u, "i^*_\alpha"] \\
K^{p-1}[L_q \cup N_p, L_p \cup (L_q \setminus \mathring{N}_p)] \arrow[u, "i^*"] \arrow[r, Rightarrow] & K^p[S_1 \wedge L_q \cup N_p, S_1 \wedge L_p \cup (L_q \setminus \mathring{N}_p)] \arrow[u, "i^*_\beta"] &
\end{tikzcd}$$

The bottom left diagramm commutes by definition. All maps with names $i^*_{something}$ are induced from obvious inclusions. The other maps are defined as follows, where p denotes the choosen point of our pointed spaces:

$$\alpha : N_q \cup N_p \cup C_1(L_q \cup N_p) \cup C_2(N_q \cup N_p) \to S_1 \wedge (L_q \cup N_p)$$
$$x \mapsto \begin{cases} p & \text{if } x \in C_2(N_q \cup N_p) \\ x & \text{else} \end{cases}$$

This map is well defined and continouos, as every point in $S_1 \wedge (L_q \cup N_p)$ is of the form $[(t_1, x)]$ where $x \in L_q \cup N_p$. and those are the points in the domain of α that not get collapsed to p We have the maps:

$$c_\gamma : [N_q \cup N_p \cup C_1(L_q \cup N_p) \cup C_2(N_q \cup N_p)] \to \left[\frac{N_q \cup N_p \cup C_1(L_q \cup N_p) \cup C_2(N_q \cup N_p)}{C_2(N_q \cup N_p)} \Big/ \frac{C_1(L_p \cup \overline{(L_q \setminus N_p)}) \cup C_2(N_q \cup N_p)}{C_2(N_q \cup N_p)} \right]$$
$$x \mapsto \begin{cases} p & \text{if } x \in C_2(N_q \cup N_p) \quad \cup \quad C_1(L_p \cup \overline{(L_q \setminus N_p)}) \\ x & \text{else} \end{cases}$$

and

$$c_\beta : [S_1 \wedge (L_q \cup N_p)] \to \left[S_1 \wedge L_q \cup N_p \Big/ S_1 \wedge L_p \cup (L_q \setminus \mathring{N}_p) \right]$$
$$x \mapsto \begin{cases} p & \text{if } x \in S_1 \wedge L_p \cup (L_q \setminus \mathring{N}_p) \\ x & \text{else} \end{cases}$$

This map is again well defined and continouos and is also a map of pairs. In fact, both maps α and β can be written as $x \mapsto \overline{x}$ and since the collapsed space is contractible, they induce isomorphism between the K-groups. The map γ is just the identity.

Furthermore, we have the commutative diagramm:

$$\begin{tikzcd} K^p[S_1 \wedge (L_q \cup N_p)] \arrow[r, "\alpha^*"] & K^p[N_q \cup N_p \cup C_1(L_q \cup N_p) \cup C_2(N_q \cup N_p)] \\ K^p\left[S_1 \wedge L_q \cup N_p \Big/ S_1 \wedge L_p \cup (L_q \setminus \mathring{N}_p) \right] \arrow[u, "c_\beta^*"] \arrow[r, "\gamma^*"] & K^p\left[\frac{N_q \cup N_p \cup C_1(L_q \cup N_p) \cup C_2(N_q \cup N_p)}{C_2(N_q \cup N_p)} \Big/ \frac{C_1(L_p \cup \overline{(L_q \setminus N_p)}) \cup C_2(N_q \cup N_p)}{C_2(N_q \cup N_p)} \right] \arrow[u, "c_\gamma^*"] \end{tikzcd}$$

To see the commutativitie we have to show that

$$c_\beta \circ \alpha = \gamma \circ c_\gamma : N_q \cup N_p \cup C_1(L_q \cup N_p) \cup C_2(N_q \cup N_p) \to \left[S_1 \wedge L_q \cup N_p \Big/ S_1 \wedge L_p \cup (L_q \setminus \mathring{N}_p) \right]$$

since α and β are both of the form $x \mapsto \overline{x} = x$ or p we compare the case distinctions:

- $c_\beta \circ \alpha(x) = p \Leftrightarrow x \in C_1\left(L_p \cup (L_q \setminus \mathring{N}_p)\right) \cup C_2(N_q \cup N_p)$
- $\gamma \circ c_\gamma(x) = p \Leftrightarrow x \in C_2(N_q \cup N_p) \cup C_1(L_p \cup \overline{(L_q \setminus N_p)}) \cup C_2(N_q \cup N_p)$

Which are the same, since $L_q \setminus \mathring{N}_p = \overline{L_q \setminus N_p}$ due to L_q being closed. Now we consider the maps:

$$c : [N_q \cup N_p \cup C_1(L_q \cup N_p)] \to \left[\frac{N_q \cup N_p \cup C_1(L_q \cup N_p)}{\mathrm{cl}\left(N_q \cup N_p \cup C_1(L_q \cup N_p) \setminus \bigcup_{j=1}^{l} C_1 V_j \right)} \right]$$
$$x \mapsto \begin{cases} p & \text{if } x \in \mathrm{cl}(N_q \cup N_p \cup C_1(L_q \cup N_p) \setminus \bigcup_{j=1}^{l} C_1 V_j) \\ x & \text{else} \end{cases}$$

and

$$
\begin{aligned}
c_\iota : [N_q \cup N_p \cup C_1(L_q \cup N_p)] &\to \left[\frac{N_q \cup N_p \cup C_1(L_q \cup N_p) \cup C_2(N_q \cup N_p)}{C_2(N_q \cup N_p)} \middle/ \frac{N_q \cup N_p \cup C_1\left(L_p \cup \mathrm{cl}\left(L_q \setminus \bigcup_{j=1}^{l} V_j\right)\right) \cup C_2(N_q \cup N_p)}{C_2(N_q \cup N_p)} \right] \\
x &\mapsto \begin{cases} p & \text{if } x \in N_q \cup N_p \cup C_1\left(L_p \cup \mathrm{cl}\left(L_q \setminus \bigcup_{j=1}^{l} V_j\right)\right) \cup C_2(N_q \cup N_p) \\ x & \text{else} \end{cases} .
\end{aligned}
$$

We define V_j to be the the finitly many components $N_p \cap S(q \to)$ and hence we can identify the sets $\overline{L_q \setminus N_p} = \mathrm{cl}(L_q \setminus (L_q \cap N_q)) = \mathrm{cl}(L_q \setminus (\bigcup_{j=1}^{l}))$ and with this the map η in the diagramm below is just the identity:

$$
\begin{tikzcd}
K^p[N_q \cup N_p \cup /(L_q \cup N_p)] & \\
K^p[N_q \cup N_p \cup C_1(L_q \cup N_p)] \arrow[u, "(m^*)^{-1}"] & K^p\left[\frac{N_q \cup N_p \cup C_1(L_q \cup N_p)}{\mathrm{cl}\left(N_q \cup N_p \cup C_1(L_q \cup N_p) \setminus \bigcup_{j=1}^{l} C_1 V_j\right)}\right] \arrow[l, "c^*"'] \\
K^p[N_q \cup N_p \cup C_1(L_q \cup N_p) \cup C_2(N_q \cup N_p)] \arrow[u, "i_\alpha^*"] & K^p\left[\frac{C_1(L_q \cup N_p)}{\mathrm{cl}\left(C_1(L_q \cup N_p) \setminus \bigcup_{j=1}^{l} C_1 V_j\right)}\right] \arrow[u, "i_\delta^*"] \\
 & K^p\left[\frac{C_1(L_q)}{\mathrm{cl}\left(C_1(L_q) \setminus \bigcup_{j=1}^{l} C_1 V_j\right)}\right] \arrow[u, "i_\eta^*"] \\
K^p\left[\frac{N_q \cup N_p \cup C_1(L_q \cup N_p) \cup C_2(N_q \cup N_p)}{C_2(N_q \cup N_p)} \middle/ \frac{C_1(L_p \cup \overline{(L_q \setminus N_p)}) \cup C_2(N_q \cup N_p)}{C_2(N_q \cup N_p)}\right] \arrow[uu, "c_\gamma^*"] & K^p\left[\frac{N_q \cup N_p \cup C_1(L_q \cup N_p) \cup C_2(N_q \cup N_p)}{C_2(N_q \cup N_p)} \middle/ \frac{N_q \cup N_p \cup C_1\left(L_p \cup \mathrm{cl}\left(L_q \setminus \bigcup_{j=1}^{l} V_j\right)\right) \cup C_2(N_q \cup N_p)}{C_2(N_q \cup N_p)}\right] \arrow[l, "\eta^*"'] \arrow[u, "c_\iota^*"]
\end{tikzcd}
$$

Now the mamoth task is to show that this beast commutes: For this we preoceed with the same strategy since again all maps are inclusions and collases and hence the maps look like $x \mapsto \overline{x} = x$ or p. For the consideration assume that $x \neq p$ in the domain. Then:

$$
\eta \circ c_\gamma \circ i_\alpha(x) = p \Leftrightarrow x \in C_1(L_p \cup \overline{(L_q \setminus N_p)}) \cup C_2(N_q \cup N_p)
$$

On the other hand:

$$c_\iota \circ i_\eta \circ i_\delta \circ c(x) = p \Leftrightarrow c(x) = p \text{ or } c_\iota(x) = p$$

$$x \in \mathrm{cl}\left(N_q \cup N_p \cup C_1(L_q \cup N_p) \setminus \bigcup_{j=1}^{l} C_1 V_j\right) \cup N_q \cup N_p \cup C_1(L_p \cup \mathrm{cl}\left(L_q \setminus \bigcup_{j=1}^{l} V_j\right)) \cup C_2(N_q \cup N_p)$$

$$= C_1 \,\mathrm{cl}\left((L_q \cup N_p) \setminus \bigcup_{j=1}^{l} V_j\right) \cup C_1(L_p \cup \mathrm{cl}\left(L_q \setminus \bigcup_{j=1}^{l} V_j\right)) \cup C_2(N_q \cup N_p)$$

$$= C_1 \,\mathrm{cl}\left((L_q) \setminus \bigcup_{j=1}^{l} V_j\right) \cup C_1(L_p \cup \mathrm{cl}\left(L_q \setminus \bigcup_{j=1}^{l} V_j\right)) \cup C_2(N_q \cup N_p)$$

$$= C_1(L_p \cup \mathrm{cl}\left(L_q \setminus \bigcup_{j=1}^{l} V_j\right)) \cup C_2(N_q \cup N_p)$$

And now the conditions agree and hence the big diagramm commutes. Now notice how we have the homeomorphism

$$\Lambda : \left[\frac{C_1(L_q)}{\mathrm{cl}\left(C_1(L_q) \setminus \bigcup_{j=1}^{l} C_1 V_j\right)}\right] \to \bigvee_{j=1}^{l} C_1 V_j \big/ \partial C_1 V_j$$

that is the identity on all $C_1 \mathring{V}_j$. Furthermore we have the compression maps

$$c_j : \bigvee_{j=1}^{l} C_1 V_j \big/ \partial C_1 V_j \to C_1 V_j \big/ \partial C_1 V_j$$

Hence by the additivity of $K-$Theorie, the pair

$$\left(K^p\left[\frac{C_1(L_q)}{\mathrm{cl}\left(C_1(L_q) \setminus \bigcup_{j=1}^{l} C_1 V_j\right)}\right], \left((c_j \circ \Lambda)^*\right)_j\right)$$

satisfies the coproduct universal property.

So consider the map

$$c^* \circ i_\delta^* \circ i_\eta^* \circ (c_j \circ \Lambda)^* : C_1 V_j \big/ \partial C_1 V_j \to K^p\left[N_q \cup N_p \cup C_1(L_q \cup N_p)\right] \tag{6}$$

This map is induced from

$$N_q \cup N_p \cup C_1(L_q \cup N_p) \to C_1 V_j \big/ \partial C_1 V_j$$
$$x \mapsto \begin{cases} x & \text{if } x \in C_1(\mathring{V}_j)\,, \\ p & \text{else }. \end{cases}$$

For all C_1V_j we need an homeomorphism to S^{k+1}:

$$r : C_1V_j\big/\partial C_1V_j \to I \times D^k\big/\partial(I \times D^k)$$
$$(t, x) \mapsto (t, \psi(x))$$

For this we use that N_p is a tubular neighbourhood of $W(\to p) \cap M^c$ where that latter is contractable. Hence, we have a homotopy equivalence by contracting the base space and the fibers simultaniouosly via a frame. and the transversality ensures, that this gives a homeomorphism from V_j to the fibre over p, which is a ball $B \subseteq W(p \to)$. and composed with the refined Morse chart this is a Ball B^{λ_p} in $\mathbb{R}^{\lambda_p}$. This gives us a well defined homeomorphism:

$$r : C_1V_j\big/\partial(C_1V_j) \to C_1B^{\lambda_p}\big/\partial(C_1B^{\lambda_p})$$

Hence, an orientation on $TW(p \to)$ induces a generator α of $K^p\left[C_1V_j\big/\partial(C_1V_j)\right]$.

On the other hand, we have a generator on $N_p \cup N_p \cup C_1(L_p \cup N_p)$ induced from an orientation on $TW(q \to)$ as follows: First we get a generator of the K group of the Regular index pair $(N_q\big/L_q) \simeq (N_q \cup N_p/L_q \cup N_p)$ by choosing a small index pair and the homotopy. This generator induces now a generator β of $K^p\left[N_q \cup N_p \cup C_1(L_q \cup N_p)\right]$.

Now we need to study, how the image of α under the map 6 relates to β: